

von DosePilot

24 response summaries from 64 traceable treatment wells

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Team composition: **Joseph Ayanda — AI/ML, software engineering, reproducibility, and experimental-design workflow.**

Public demo video: <https://youtu.be/QeOGJlgx378> (75 seconds)

DosePilot freezes a physical assay layout before responses arrive, validates every purchased drug-dose-plate identity, and reconstructs 24 fixed research summaries with an explicit missing-data and evidence contract.

Current verified development result	Value
Patient-balanced full-24 MSE	0.001042745722
Improvement versus bandwidth-0.7	1.4674%
Improvement versus original R13	8.9193%
Patient wins versus bandwidth-0.7	40 / 59
Favorable folds versus bandwidth-0.7	5 / 5
p90 patient RMSE	0.037419696
Physical treatment wells	64 per deployment

The orientation-specific control-quality rank-1 model is the **lowest verified Lib1 development point estimate** currently recorded. It is repeated adaptive development, not independent confirmation, an official competition score, or a clinical result.

Repository: <https://github.com/josepha-mayo/von-dosepilot>

Live fictional-data demonstration: <https://von-dosepilot.netlify.app>

Executive summary

The decision DosePilot makes

Sparse screening is a physical measurement problem before it is a prediction problem. Each feature corresponds to a drug, an exact dose, a source plate, and a treatment well. DosePilot commits a 64-treatment plan before responses arrive, then reconstructs 24 normalized log-dose AUC summaries.

The development contract is deliberately strict:

- 119 organoid samples are grouped into 59 whole patients.
- The endpoint is fixed for all 24 drugs.
- Each A or B deployment uses exactly 64 distinct treatment wells, 32 per source plate.
- Every sample from one patient remains on the same side of inner and outer splits.
- A and B are scored separately; their losses are averaged.
- A/B prediction vectors are never averaged into a hidden 128-well ensemble.
- Planning, scaling, fitting, and inner spectral selection stay inside fitting splits.

What changed on 6 October

The current candidate preserves the 64-well treatment acquisition and the bandwidth-0.7 residual geometry, then uses standard plate-control summaries to estimate a small shared target-deviation mode.

Orientation B uses an 11-column quality basis containing control range, negative/positive control level, and negative/positive control variation. Orientation A uses a more conservative seven-column level basis that omits the CV channels. Both corrections are rank 1 and deliberately damped.

The result is **0.001042745722 MSE**, 1.4674% below bandwidth-0.7 and 8.9193% below R13. Against bandwidth-0.7 it wins 40/59 patients, all 5/5 outer folds, p90, and 19/24 target-average errors.

What is not demonstrated

The same 59-patient development population has been reused across many research rounds. Patient grouping prevents within-run patient leakage, but repeated inspection means the current point estimate is not untouched validation.

The immediate predecessor is already very close: 0.001043179589. The current candidate is only 0.0416% lower and wins 3/5 folds versus that immediate predecessor. The report therefore does not claim uniform incremental superiority.

Protected22 is closed and exposed; its full primary was not estimable. No Protected22 response was used to develop or verify the current candidate.

Endpoint and physical budget

Fixed response target

Each of the 24 outputs is the arithmetic mean of two identified source-plate trapezoidal AUCs over log concentration. Supplied normalized viability values are used without clipping.

The output is not IC50, a drug rank, or a clinical-response label.

Sixty-four treatment wells

The retrospective source profile contains 416 eligible target-treatment measurements per sample across two plates. The frozen plan selects two native doses for every target and allocates a third-dose upgrade to 16 targets:

$$24 \times 2 + 16 \times 1 = 64 \text{ treatment wells}$$

Each alternative deployment has 32 p1 and 32 p2 treatment wells. Standard vehicle/viability controls are separate assay resources and are not counted as treatment wells.

The 64/416 ratio is a **treatment-measurement count** comparison. It is not a demonstrated 84.62% reduction in real laboratory cost, material, or time.

Frozen schedule property	A	B
Treatment wells	64	64
p1 treatment wells	32	32
p2 treatment wells	32	32
Two-dose targets	8	8
Three-dose targets	16	16

A and B are complete alternative 64-well layouts. Their prediction vectors are never combined.

Data sources, licenses, and compliance

The primary development source is the public colorectal-cancer patient-derived organoid study of **Kryeziu et al., Cell Reports Medicine (2026)**. DosePilot pins the public **Mendeley Data v3 Data S4.xlsx** workbook at 15,886,254 bytes with SHA-256 `3847aa93b2a84c7d5d0b04c26494f39f35963fc41e96eae97d8a180fbc33d81c`. The deposit is listed under **CC BY 4.0**. The public reconstruction route selects the historical Lib1 TRAIN measurements and records zero Lib2 numerical-response conversions and zero raw-signal conversions.

Original DosePilot code, documentation, and fictional fixtures are released under **MIT**. External adaptation and stress-test datasets retain their own source licenses and are documented separately. The public repository excludes biological workbooks, patient-level prediction arrays, fitted biological weights, and protected responses. Software dependencies are declared in `requirements.txt` / `study/requirements.txt` and retain their upstream licenses.

No current accuracy experiment uses a paid model API, proprietary hardware requirement, or private clinical dataset. The historical fitted arrays used for exact replay remain private; the public source-to-results route reconstructs the supported development pipeline from the cited public source.

Model

Stage 1 - training-contained acquisition

For each drug, the allocator compares two-dose and three-dose native subsets with a fitting-only covariance criterion. Sixteen targets receive a third-dose upgrade. Acquisition is recomputed inside fitting splits.

Stage 2 - own-drug ridge

Each target has an unpenalized-intercept ridge head that sees only the purchased measurements for that drug. One shared ridge penalty is selected inside patient-grouped inner folds.

Stage 3 - cross-drug residual correction

The bandwidth-0.7 additive model fits residual structure with grouped Gaussian kernels plus a linear component. Ten residual spectral options are evaluated in inner patient folds.

Stage 4 - control-quality rank-1 calibration

The current candidate uses standard plate-control summaries, not additional treatment wells.

Orientation B quality basis (11 features):

- control range mean and half-difference;
- squared range mean;
- negative-control log-median mean and half-difference;
- positive-control log-median mean and half-difference;
- negative-control CV mean and half-difference;
- positive-control CV mean and half-difference.

A fitting-only ridge regression is reduced to its leading target-deviation mode, then applied at replacement strength **1/3**.

Orientation A level basis (7 features):

The same range and control-level features are used, but the CV channels are omitted. The rank-1 replacement strength is **1/9**.

The asymmetry is intentional. Orientation A's richer quality basis was noisier in crossfit diagnostics, while B benefited from the full quality basis.

Rank 1 is also intentional. Recorded rank-depth diagnostics show ranks 2, 3, and 4 all worsen the held-patient MSE.

Same-task development results

Complete procedure	Patient-balanced MSE	Interpretation
R13 own-drug reconstruction	0.001144858681	historical baseline
R18 exploratory candidate	0.001141404811	historical reference
S2 spectral residual	0.001070143945	superseded
Additive group kernel, bandwidth 1.0	0.001060552730	predecessor
Bandwidth-0.7 additive	0.001058275042	public operational baseline
Standard-control meanquad/plate interaction v2	0.001045585078	adaptive development
Control rank-1 target deviation	0.001044448281	adaptive development
Orientation-A auxiliary rank-1	0.001043179589	immediate predecessor
Orientation-specific control-quality rank-1	0.001042745722	lowest verified development point estimate

Breadth versus bandwidth-0.7

Metric	Current	Bandwidth-0.7
MSE	0.001042745722	0.001058275042
p90 patient RMSE	0.037419696	0.037894285
Orientation A MSE	0.001100608798	0.001104752162
Orientation B MSE	0.000984882646	0.001011797922

The current model wins **40/59 patient means**, all **5/5 outer folds**, and **19/24 target-average errors** versus bandwidth-0.7.

The five regressing target averages are Afatinib, AZD7762, LCL161, Regorafenib, and Trametinib.

Descriptive patient uncertainty

A post-hoc **100,000-resample whole-patient bootstrap** (seed 20261006) resamples the 59 patient-level loss differences without rerunning model selection. For current-minus-bandwidth-0.7 mean loss, the descriptive percentile 95% interval is **[-2.60e-5, -6.54e-6]**, entirely below zero. This interval is **selection-naïve** because Lib1 has been repeatedly inspected; it is not a confirmatory or selection-corrected confidence interval.

The aggregate-only receipt is `evidence/current_successor_descriptive_bootstrap_20261006.json` ; no patient-level rows are published.

Historical screens

Versus R13: **8.9193% lower MSE, 49/59 patient wins, 5/5 folds, 22/24 target-average wins.**

Versus authenticated R18: **8.6437% lower MSE, 50/59 patient wins, 5/5 folds, p90 nonworse**, and both orientation-wide MSEs remain below the reference expected MSE.

Immediate-predecessor honesty check

Versus the 0.001043179589 immediate predecessor, the current point estimate is only **0.0416% lower**, with **32/59 patient wins, 27 losses, and 3/5 favorable folds**. p90 remains nonworse. The same descriptive patient bootstrap gives **[-1.13e-6, 2.58e-7]**, which crosses zero. This is why the current model is described as the lowest verified adaptive-development point estimate, not as uniformly superior to every nearby candidate.

Falsification and verification

Exact replay

The frozen verifier reports:

Check	Result
Status	PASS_FLOATING_REPLAY
Base model refit	false
Calibration recomputed	true
Held-target predictions reconstructed	5,712
Maximum prediction difference	2.22e-16 (fresh Windows replay)
Prediction tolerance	5e-16
Calibration folds checked	5
Protected22 access	false

The saved candidate prediction SHA-256 is bound in the evidence receipt.

Extensions that failed

The project keeps negative branches visible.

- rank-2, rank-3, rank-4 control-quality models: all worse than rank 1;
- robustized quality bases: worse;
- target-gated quality corrections: worse;
- wider bandwidth 1.4: rejected;
- cross-patient median bandwidth: lower point mean but only 4/5 folds, rejected;
- simplex spectral stacking: all inner objectives improved but held-patient mean/breadth/tail worsened;
- isotonic paid features: rejected.

This matters because repeated development can manufacture attractive decimals. DosePilot keeps the failed gates and does not splice targets or folds after seeing outcomes.

External evidence and prospective boundary

Protected22

The frozen Lib2 primary required 19,642 cells across 61 PDOs, 31 patients, and 22 targets. Five required cells were unavailable, so the full primary is **NOT_ESTIMABLE**. A conditional complete-patient diagnostic exists, but it is not the primary, and prior exposure prevents an untouched-confirmation claim.

All 61 PDOs and 31 patients are treated as exposed. No Protected22 value was used to build the current Lib1 successor.

Separate external adaptations

A matched-CAF design adaptation passed its own frozen gate on a different cohort and endpoint panel. FORECAST-1 and eLife adaptations did not fully pass their preset gates.

These are design-level external studies. They do not directly validate the current fitted 24-output Lib1 weights.

Correct next experiment

The scientifically strong next step is not another adaptive Lib1 sweep. It is a preregistered, patient-separated prospective or untouched assessment in which device identity, QC, missingness, comparator, output denominator, and success gates are frozen before responses are collected.

Reproducibility and software

The repository separates three kinds of evidence:

1. **Biological development evidence:** grouped held-patient results and aggregate receipts.
2. **Numerical reproducibility:** frozen replay of saved candidate predictions without refitting the base model.
3. **Operational software:** an identity-bound fictional lifecycle that demonstrates commit, missing-input withholding, recovery, and prediction.

Current candidate artifacts:

```
evidence/orientation_specific_control_quality_rank1_20261006.json
evidence/current_successor_descriptive_bootstrap_20261006.json
study/orientation_specific_control_quality_rank1/PROPOSAL.json
study/orientation_specific_control_quality_rank1/PROTOCOL.json
study/orientation_specific_control_quality_rank1/FREEZE.json
study/orientation_specific_control_quality_rank1/run_study.py
study/orientation_specific_control_quality_rank1/verify_study.py
```

The existing durable fictional lifecycle still represents the bandwidth-0.7 operational baseline. The newer calibration layer has not been silently substituted into that older demo. The separation is explicit so a reviewer can distinguish an accuracy experiment from a completed product-runtime migration.

The public repository excludes patient arrays, fitted biological weights, and protected responses.

AI assistance and runtime independence

ChatGPT assisted with research synthesis, implementation, numerical checking, adversarial review, and documentation. The DosePilot reconstruction runtime itself uses **no language-model API or paid model service**. AI assistance does not replace the frozen numerical verifiers, patient-grouped evaluation, source hashes, or preserved negative results.

Claim boundary

DosePilot currently supports:

- a fixed 64-treatment acquisition contract;
- a lowest verified Lib1 adaptive-development MSE of 0.001042745722;
- broad improvement versus bandwidth-0.7;
- exact frozen numerical replay;
- transparent adverse slices and negative experiments;
- an executable fictional operational workflow;
- a clear prospective handoff.

It does **not** establish clinical utility, realized laboratory savings, prospective organ-on-chip performance, independent validation of the current fitted weights, an official competition score, or a guaranteed outcome.

References

1. Kryeziu et al., *Cell Reports Medicine* (2026), DOI: 10.1016/j.xcrm.2026.102840.

2. Tan et al., *Cell Reports Medicine* (2023), DOI: 10.1016/j.xcrm.2023.101335.
3. Farin et al., *Cancer Discovery* (2023), DOI: 10.1158/2159-8290.CD-23-0050; Mendeley Data DOI: 10.17632/fypp6xhkjy.1.
4. Verissimo et al., *eLife* (2016), DOI: 10.7554/eLife.18489.